

Amit Maish

amit.maish1@gmail.com — [github:amitmaish](https://github.com/amitmaish) — (818) 860-6459

I am a student at CalArts seeking an internship in IT or Digital Signal Processing that will build on my deep and active interest in software engineering and computer technology. Through many self-guided projects that apply technology to music and art I have become proficient in Linux and MacOS system administration and software development in Rust, C/C++, and Javascript. I look forward to using and advancing my experience in a professional capacity.

Projects

Because of my interests in computers, I've taken extracurriculars in CS as well as written a few tools to assist in my music production & sound exploration:

Audio Manipulation with the Fast Fourier Transform:

I used an FFT library and JUCE to create a new type of distortion effect. It works by running the audio through the FFT and selecting the N most prominent frequencies. It then mutes everything else before running it through the inverse FFT to bring up back into the time domain. This creates a uniquely digital form of distortion. I'm now working on the GUI as well as adding more options for the FFT distortion.

Game of Colors:

Designed a version of Conway's Game of Life that's generalized to multiple dimensions. In order to support N dimensions I generalized the math to use vector operations. The program is implemented in Rust and the frames were combined into a video using ffmpeg. I ran a popular video through the algorithm: https://youtu.be/ZK_-tXo9KA0?si=5k5TbYx53J7GLOF0

Custom System Configuration and Management

I manage multiple MacOS and Linux computers and need a reliable way to synchronize changes across them. This lead me to develop my own declarative system to perform distributed configuration management for all of my computers from a single set of configuration files to make this as seamless as possible. I designed the system to use git to ensure that any changes are easily tracked.

Doodles

For my final in a javascript art class I implemented a custom vector graphics solution to emulate the look of pencil on paper. I drew bezier curves as a series of points. Then at runtime I generated a series of points that look like hand drawn flowers. The end result is a procedural doodle generator that dynamically draws a new bed of flowers whenever you reload the page. <https://amitmaish.github.io/coding-art-fundamentals/doodle>

Other Jobs

Audio Engineer 2018 to 2025

- Served as the primary audio engineer for live performances at my High School and Moorpark College
- Was responsible for designing the audio system we used
- Performed maintenance of all the equipments
- Instructed and mentored the audio team

Bigger Hammer 2022 to 2025

- Setup and tear down of audiovisual equipment for music festivals and other clients across LA.

Education

AA: Moorpark College 2020 to 2024

BFA: CalArts 2024 to 2026